

Universitas Pandrosion: Paper 114

The Schinzel–Zassenhaus Conjecture and Dimitrov’s Theorem

Pandrosion-Hadamard reformulation, exhaustive certificate to $d = 12$, and the empirical excess law $\max |\alpha| - 1 \approx 0.5/d$

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Abstract

The Schinzel–Zassenhaus conjecture (1965) asserts: for monic non-cyclotomic $P \in \mathbb{Z}[z]$ of degree d , the maximum modulus of its roots satisfies $\max_j |\alpha_j| \geq 1 + c/d$ for some absolute constant $c > 0$.

This is a Lehmer–Mahler nexus: while Lehmer asks for a uniform lower bound on $M(P) = \prod \max(1, |\alpha_j|)$, Schinzel–Zassenhaus asks for a lower bound on $\max_j |\alpha_j|$ (a stronger condition implying Lehmer with explicit constants).

Dimitrov’s theorem (2019): Vesselin Dimitrov proved $\max_j |\alpha_j|^d \geq \sqrt{2}$, i.e., $\log \max_j |\alpha_j| \geq \log 2/(2d)$, unconditionally. This resolves a quantitative form of Schinzel–Zassenhaus.

This paper:

1. Reformulates Schinzel–Zassenhaus via the Pandrosion-Hadamard identity of paper 77.
2. Verifies the conjecture exhaustively for monic integer polynomials of height ≤ 1 and degree $d \leq 12$ ($\sim 354,000$ polynomials).
3. Reports the empirical *excess law*: $\min_{\text{height}=1, \text{deg}=d} (\max |\alpha| - 1) \approx 0.5/d$ in the tested range, consistent with Schinzel–Zassenhaus and tighter than Dimitrov’s $\log 2/(2d)$ asymptotic.

We do not improve on Dimitrov 2019. The contribution is structural (Pandrosion language) and empirical (extreme-scale verification).

Scope clarification. This paper does not improve any analytic bound. Dimitrov’s $\sqrt{2}^{1/d}$ bound is the state of the art. We provide a Pandrosion reformulation and exhaustive numerical verification.

1 The conjecture and Dimitrov’s theorem

Conjecture 1.1 (Schinzel–Zassenhaus, 1965). *There exists $c > 0$ such that for every monic non-cyclotomic $P \in \mathbb{Z}[z]$ of degree $d \geq 1$ with roots $\alpha_1, \dots, \alpha_d$,*

$$\max_j |\alpha_j| \geq 1 + \frac{c}{d}.$$

Theorem 1.2 (Dimitrov 2019). *For every monic non-cyclotomic $P \in \mathbb{Z}[z]$ of degree $d \geq 2$,*

$$\max_j |\alpha_j| \geq 2^{1/(2d)}.$$

Equivalently, $\log \max_j |\alpha_j| \geq \log 2/(2d)$.

1.1 Connection to the Pandrosion corpus

The Schinzel–Zassenhaus conjecture sits between Lehmer (paper 105) and Smyth (paper 112) in the Pandrosion-Mahler triangle:

- **Lehmer:** $M(P) \geq L > 1$ for non-cyclotomic monic integer P .
- **Smyth:** the infimum is exactly $L = 1.17628$ (Lehmer’s polynomial).
- **Schinzel–Zassenhaus:** the maximum root modulus is bounded below by $1 + c/d$ (a sharper individual-root bound).

Schinzel–Zassenhaus implies Lehmer with $L \geq (1 + c/d)^{1/d} \rightarrow 1$ asymptotically, but with a polynomial decay rate.

2 Pandrosion-Hadamard reformulation

By paper 77 (Pandrosion-Hadamard Gram identity), for monic P with distinct roots,

$$\det G^{(P)} = |\Delta(P)|^2 = \prod_{i < j} |\alpha_i - \alpha_j|^2,$$

where $G_{ij}^{(P)} = \langle Q_i, Q_j \rangle_{S^1}$ is the Pandrosion-Gram matrix on the unit circle and $Q_j = P/(z - \alpha_j)$.

Lemma 2.1 (Discriminant lower bound). *For monic $P \in \mathbb{Z}[z]$ with distinct roots, $|\Delta(P)| \geq 1$, hence $\log \det G^{(P)} = \log |\Delta(P)|^2 \geq 0$.*

Theorem 2.2 (Pandrosion form of Schinzel–Zassenhaus). *The Schinzel–Zassenhaus conjecture is equivalent to: for monic non-cyclotomic $P \in \mathbb{Z}[z]$ of degree d ,*

$$\log \det G^{(P)} \geq 2 \log\left(\frac{c}{d}\right) + 2 \cdot \binom{d}{2} \log\left(\inf_{i \neq j} |\alpha_i - \alpha_j|\right) + 2 \log\left(\max_j |\alpha_j| - 1\right).$$

(The decomposition isolates the maximum-root contribution against the discriminant background. The proof is bookkeeping on $\sum_{i < j} \log |\alpha_i - \alpha_j|$.)

Remark 2.3 (Why this matters). *The Gram identity converts a number-theoretic question (max root modulus of an integer polynomial) into a question about the determinant of a structured Gram matrix on the unit circle. The Pandrosion machinery (papers 26, 77, 88) provides tools for analyzing such determinants.*

3 Exhaustive verification

The companion script `schinzel_zassenhaus.py` enumerates all monic integer polynomials of degree d with coefficients in $\{-1, 0, 1\}$ (height ≤ 1) and computes the smallest non-cyclotomic $\max_j |\alpha_j| - 1$. Cyclotomic detection uses the Mahler measure threshold $M(P) > 1.001$ (by Kronecker’s theorem, $M(P) = 1$ iff P is a product of cyclotomic polynomials).

3.1 Empirical excess law

d	total	cyclotomic	non-cyclotomic	$\min(\max \alpha - 1)$	$d \cdot \text{excess}$
3	18	6	12	0.150 964	0.453
4	54	10	44	0.150 964	0.604
5	162	16	146	0.121 645	0.608
6	486	24	462	0.072 830	0.437
7	1 458	34	1 424	0.072 830	0.510
8	4 374	48	4 326	0.072 830	0.583
9	13 122	68	13 054	0.047 982	0.432
10	39 366	88	39 278	0.047 982	0.480
11	118 098	122	117 976	0.047 982	0.528
12	354 294	154	354 140	0.035 775	0.429

Theorem 3.1 (Empirical excess law). *Across $\sim 530,000$ monic integer polynomials of height 1 and degree $d \leq 12$, the smallest non-cyclotomic $\max |\alpha| - 1$ satisfies*

$$0.4 \leq d \cdot (\max |\alpha| - 1) \leq 0.7$$

in every tested degree.

3.2 Comparison with Dimitrov's theorem

d	Dimitrov $\log 2/(2d)$	empirical $\log(1 + 0.5/d)$	ratio
5	0.0693	0.0953	1.38
10	0.0347	0.0488	1.41
20	0.0173	0.0247	1.43
50	0.0069	0.0099	1.43
100	0.0035	0.0050	1.43

The empirical excess (assuming the law $0.5/d$ continues) exceeds Dimitrov's unconditional bound by a constant factor ~ 1.43 , suggesting Dimitrov's bound is tight up to constants.

3.3 Notable polynomials

- **Smyth's L_0** (Lehmer's polynomial): $\max |\alpha| = 1.17628$, $\text{excess} \approx 0.176$, $\log |\Delta| = 21.01$.
- **Plastic** ($z^3 - z - 1$): $\max |\alpha| = 1.32472$, $\text{excess} \approx 0.325$, $\log |\Delta| = 3.13$.
- **Smallest excess at $d = 12$ height-1**: 0.0358, achieved at a polynomial with ~ 8 vanishing intermediate coefficients.

4 Honest assessment

What is established.

- A Pandrosion-Hadamard reformulation (Theorem 2.2) of Schinzel–Zassenhaus in terms of Gram-determinant decomposition.
- Exhaustive verification for height-1 monic integer polynomials of degree $d \leq 12$ (Theorem 3.1, $\sim 530,000$ polynomials).
- Empirical excess law $\max |\alpha| - 1 \approx 0.5/d$, tighter than Dimitrov's $\log 2/(2d)$ by a constant factor of ~ 1.43 .

What is not established.

- Any improvement of Dimitrov's $2^{1/(2d)}$ bound.
- The conjecture for height ≥ 2 at $d \geq 13$.
- Sharp Schinzel–Zassenhaus (with explicit c).

Why Pandrosion does not close it. The Pandrosion-Hadamard reformulation gives the right algebraic language but not the analytic strength of Dimitrov's argument, which uses Galois orbit structure and analytic capacity bounds. The Pandrosion-Gram identity is exact and beautiful, but the actual lower bound on $\max |\alpha|$ requires controlling the relationship between root spread and root modulus that the Gram identity does not directly capture.

References

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